

# CryoTrack: Blockchain India Challenge Submission

## 1. Problem Statement

India's supply chain infrastructure is highly fragmented from farmers, warehouse operators to transporters and distributors. Only around 6% of the market is digitised currently which shows how the current solutions are lacking India's needs. India's market is unique with more than 90% of logistics unorganised, retail market consisting of 1 to 1.5 Crore locally owned Kiranas and shops remain unorganised and the most hit agricultural sector faces a loss of an estimated **40% of food and agricultural production** annually due to being unorganised.

Beyond fragmentation, traditional solutions suffer from the **"Garbage In, Garbage Out" (GIGO) vulnerability** which is a core problem in the Indian market. Even when data is digitized, it relies on manual entry by logistics personnel or IOT devices which can be compromised. Based on research manual data entry carries an error rate of 1 to 5% which seems small but compounds with thousands of touchpoints in a supply chain. In Cold Storage logistics around 70% still rely on manual logging from workers, compliance often needs workers manually recording temperatures 2 to 3 times a day. This leaves **88% to 90% of the operational time without logging**. To avoid penalties or losing their jobs, workers fake the manual logs (writing down that the temperature was fine even if the power went out), feeding garbage into the system. GIGO vulnerability not just impacts financials but costs lives and is fatal. An example of this is the 2022 cough syrup tragedy in The Gambia. The supply chain for the syrup was robust and had all the correct entries but still it turned out to be a tragedy resulting in over 70 plus children deaths. Similarly the recent Coldrif tragedy which resulted in at least 24 children deaths, the 2020 Ramnagar Poisoning where 17 children died after consuming contaminated Coldbest-PC cough syrup.

Current supply chain solutions are limited due to their approach with security relying only on expensive, enterprise grade IOT devices which require professional help for setup and monitoring. This is also not cost effective and feasible considering India's logistics sector which is driven by around 6.3 crore MSMEs working on slim margins. Due to this often cheap grade unencrypted tracking alternatives are used which provide a dangerous 'false sense of security.' For example, a cheap sensor can be physically moved to an AC vent to hide a cold chain failure, or an unencrypted log can be altered due to negligence or a localized bribe. This leaves the supply chain fundamentally broken at the ground level. These vulnerabilities are not addressed with most of the current solution's preventive approach. Their data integrity is dependent on the source's honesty and can be easily compromised as seen in historical examples. These limitations, vulnerabilities and lack of digitisation of the Indian market which still stays at around 6% shows how current solutions are not fit for the Indian market needs and a new innovative approach is needed which should be scalable, affordable, secure and fast.

## 2. Innovative Solution

CryoTrack is a one stop solution for the supply chain needs of India. It's an actor and industry agnostic protocol which is designed to be scalable, fast, secure and traceable as fit for the Indian market. The core innovation lies in the custom application level consensus integrated in the HyperLedger Fabric Permissioned Blockchain which enables a light node consensus version of RAFT, agnostic stock proofs, logs and limitless integration. Additionally we focus on a mobile first approach with easy and straightforward UX integrated with edge-optimized mobile deployed and predictive AI. CryoTrack is built considering the current solutions meaning it can be integrated to replace the current solution and our innovative solution provides an extra layer of security and traceability.

The design principles we had in mind which led to this innovative solution were to achieve 100 percent traceability especially in case of malicious actors, a practical solution which can be scaled all over India from small farmers to enterprises, it should be industry agnostic meaning the same protocol should be able to serve across different industries like Pharma, Chemical, Agri etc. and lastly it should be secure and follow the government guidelines for streamlined adoption.

### 2.1 HyperLedger Fabric with custom application level consensus: The Checkpoint Mechanism

CryoTrack uses a permissioned HyperLedger chain to maintain the supply chain. HyperLedger Fabric is a clear winner for CryoTrack as it is purpose built for enterprise B2B networks. Fabric eliminates the need for mining, gas fees, and tokenomics, replacing them with a deterministic, high-throughput custom consensus mechanism. With Fabric and our consensus we can process thousands of telemetry pings and supply chain events per second at zero variable cost per transaction. Furthermore, Fabric's modular architecture seamlessly integrates with our mobile-first approach and API gateways, ensuring that the backend infrastructure remains invisible to the ground-level workforce while providing enterprise-grade immutability to stakeholders.

Our application level consensus: The Checkpoint Mechanism enables us to achieve near zero garbage inputs with 100 percent traceability based on game theory and economic principles. We shift the security structure instead of relying on just the logs and manual entries to create a self-policing, highly secure network.

### 2.2 Targeting GIGO with Game Theory: The Handoff Liability

GIGO vulnerability impacts the most when actors can log false data without immediate consequence, such as a driver manually recording a safe temperature after a power failure. We neutralize this with a strict game-theoretic model of **100% Liability Transfer** at every node in the supply chain. Within the CryoTrack protocol, when two actors of the chain interact for a physical handoff of goods (e.g., from a warehouse operator to a transporter, or a transporter to a local Kirana), this is logged as a cryptographic checkpoint on the Fabric ledger. Here, the

receiving actor must verify the state of the good and the accompanied telemetry data. The game theory implications of this approach rests in **Assumption of Liability** meaning if the actor accepts the transfer without logging a dispute or challenge with proof, they accept the goods and confirm the state presented by the other actor. They will legally and financially assume 100% liability for the state of the goods from that moment forward. In case something goes wrong, because no actor wants to bear the financial penalty for a previous actor's negligence, they are highly incentivised to audit and verify the state of the goods and data at the checkpoint. This zero-sum liability model forces honesty. Although with this we cant say we have fixed the GIGO vulnerability but the idea being the cost of collusion or accepting "garbage" data becomes inherently higher than the cost of compliance or colluding.

## **2.3 Scalability and Affordability for the Indian Market**

Our application level consensus makes CryoTrack uniquely scalable for the fragmented Indian market. Because the security is derived from protocol's game theoretic checkpoints rather than just the hardware itself, we remove the barrier of high end IOT setups for small scale users. They can become part of the ledger from our App or Website using affordable devices. The IOT or manual logs will be utilized to be used as proofs in case of disputes and help in correct resolutions. So depending on different supply chains we can ensure certain logging as mandatory or optional making CryoTrack an extremely flexible solution and shows how it fits across different supply chains. By utilizing Hyperledger Fabric's permissioned architecture we don't need gas fees and tokenomics. With our checkpoint consensus we can reduce the heavy lifting needed by the endorsement policies of the ledger making us more scalable and enable us to handle drastically more high volume, low margin operations like agriculture and local distribution. CryoTrack uses a Deferred Endorsement Policy approach building a pragmatic consensus. With this we can reduce the signatures needed in Endorsement policies to just one and make our ledger faster and scalable.

## **2.4 Resolution for disputed Checkpoints**

In the unhappy case , the receiving actor claims that the goods state is compromised and is not true in respect to the ledger state or the guidelines, indicating that the provider actor is malicious. A dispute is created, now different strategies can be used based on different supply chains following the same agnostic structure. Depending on the goods value and supply chain needs we can have different resolution mechanisms fulfilling our goal of supporting all Indian supply chains. We propose two solutions for different supply chains based on the consumer and enterprise needs. Firstly, we use special permissioned nodes which will be able to vote and decide which actor is true. These permissioned nodes will be only accessible to a set of actors which are either approved by the government of India, the government agencies we are working with like FSSAI or the enterprise implementing CryoTrack for supply chain. Secondly, we can outsource the dispute resolution to a decentralized resolution protocol like Kleros. A real world example for it, Kleros's participation with the municipal court of Lavalle in Mendoza, Argentina for small scale disputes. For even more scalability we can have a combination of both where the first decision can be outsourced to a neutral third party or AI model but the final decision will be

taken by the permissioned entity. It's important to create a neutral third party for the decision making, to ensure no actor is penalised without being at fault.

## 2.5 Two-Tier AI Architecture: Edge-Optimized and Centralized Intelligence

CryoTrack's AI is not a standalone feature in fact it is directly embedded into the Checkpoint Mechanism at two moments where data integrity matters most: when **data enters the system**, and when **liability is transferred between actors**. The protocol's security model is based on checkpoints and incentive design which only works if participants can reliably verify the truth before accepting responsibility. CryoTrack's AI acts as an **intelligent verification assistant**, augmenting human judgment rather than replacing it. It highlights anomalies, surfaces relevant context, and reduces the probability of manipulated or erroneous data entering the ledger, thus addressing the core "Garbage-In, Garbage-Out" challenge in supply chains. To achieve this, CryoTrack employs a **two-tier AI architecture**, where each tier solves a distinct problem at a specific point in the workflow.

### 2.5.1 Tier 1: The On-Device Integrity Layer

Tier 1 runs entirely on the ground-level worker's existing smartphone and is built around one constraint above all else: it must work reliably under real Indian logistics conditions, meaning low-end Android devices, intermittent connectivity, and workers with no technical background. For low-infra actors, the entire data entry process today is manual, paper-based and easily falsified. Lorry receipts, warehouse logs and condition records are handwritten, retyped or self-reported with no structural mechanism to verify their accuracy. Tier 1 addresses this directly through a lightweight data capture layer built around three mechanisms. First, on-device OCR digitises paper-based logistics documents such as lorry receipts and warehouse logs at the point of capture, replacing manual retyping with structured digital records that feed directly into the ledger. Second, guided structured input replaces free-text fields with constrained inputs wherever possible, reducing the surface area for accidental or intentional bad entries. Third, timestamped and GPS-tagged photo logging allows workers to capture the physical state of cargo, storage units and vehicles at each checkpoint, creating tamper-evident contextual evidence anchored immutably to the handoff record. Beyond data capture, Tier 1 performs fast deterministic threshold checks, verifying whether incoming telemetry falls within the SLA-defined parameters for that shipment and warning the worker immediately if a breach is detected. When operating in zero connectivity environments such as cold storage facilities or highway corridors, Tier 1 locally caches all cryptographic signatures, telemetry readings and OCR captures using Write-Once-Read-Many enforcement, automatically anchoring the batch to the Hyperledger Fabric ledger once connectivity is restored and preserving the integrity of the timeline. Tier 1's value is reliability, not sophistication. Its job is to ensure data entering the ledger is honest, complete and tamper-evident, and ready to be passed to Tier 2 the moment a liability transfer is initiated.

### 2.5.2 Tier 2: The Centralized Evaluation Engine

Tier 2 runs on CryoTrack's servers and operates in two intentionally decoupled modes so that real-time handoff evaluation is never delayed by background processing. The first is a latency-sensitive real-time path that fires the moment a liability transfer is initiated and data is received from either Tier 1 or the API integration layer. Before the receiving actor confirms acceptance, it runs a rapid multi-signal evaluation: a server-side telemetry verification against SLA parameters across the full journey trace rather than just the current reading, a pattern analysis comparing the shipment's telemetry history against expected parameters for that route and goods class, and a critical external signal correlation that cross-references telemetry against real-time weather APIs, route duration estimates and traffic conditions. This last layer is where the most important detections occur. If telemetry shows the refrigeration unit was inactive during a transit window and weather data confirms ambient temperatures during that period were sufficient to breach SLA thresholds for that goods class, Tier 2 immediately surfaces this to the receiving actor before they confirm acceptance. Every flag is surfaced as a plain-language explanation citing the exact signals that triggered it, which is the system's core application of **Explainable AI**. Instead of merely outputting an opaque, unexplainable AI risk metric, the protocol leverages Explainable AI (XAI) to translate the underlying causal anomaly into actionable plain language. CryoTrack ensures every alert is transparent, auditable and immediately actionable by ground-level workers, enterprise stakeholders and regulatory bodies alike. The actor retains full autonomy over their decision; Tier 2 exists solely to ensure that decision is informed rather than blind, and that no actor unknowingly inherits a compromised shipment. If connectivity is unavailable at the moment of handoff, the evaluation is deferred, the actor is clearly notified, the pending flag is logged on-chain and the evaluation is anchored the moment connectivity is restored. The second is an asynchronous analytics path running independently of individual handoffs, analyzing aggregated immutable network data to identify systemic patterns and risk signals. Critically, this layer applies **causal AI** rather than standard correlation-based models, identifying cause-and-effect relationships between route conditions, actor behavior and supply chain failures rather than surface-level statistical patterns. A dedicated model is trained on this data over time, continuously improving the accuracy of the real-time path as the network grows.

## 2.6 Verified Compliance Record and Regulatory Integration

As a byproduct of every checkpoint, CryoTrack maintains a **Verified Compliance Record** for each actor on the network, a structured, timestamped and on-chain history of disputes, anomaly flags and clean handoffs. This is not a public-facing score but a regulatory-grade audit resource. During inspections, license renewals or recall investigations, regulators such as FSSAI and CDSCO gain access to a verified, tamper-proof compliance history per actor rather than relying on fragmented paper logs or self-reported data, turning what is currently a weeks-long manual process into an immediate structured audit. This foundation also defines a clear Phase 2 trajectory: as the network reaches sufficient scale, the on-chain compliance history naturally evolves into a network-wide reputation system enabling marketplace-grade features such as verified counterparty selection and contract bidding. The data infrastructure being built today is the prerequisite for that layer. In line with the Digital Personal Data Protection Act 2023, all data fed into cross-actor analytics follows data minimization principles, ensuring no unnecessary personally identifiable information is retained or processed at the network level.

## 2.7 Zero-Friction UX: A Mobile-First Strategy for Last-Mile Adoption

CryoTrack is designed to serve the entire market spectrum, from small farmers and local transporters to large pharmaceutical distributors and enterprise logistics operators. For actors with existing infrastructure, onboarding happens through the API gateway, integrating directly with their ERP or warehouse management systems without requiring any new interface. For the far larger segment of low-infra actors that make up the majority of India's 6.3 crore MSME-driven logistics sector, the entry point is different. These actors have no enterprise software, no IoT infrastructure and no technical background, yet they are the most critical nodes in the supply chain, especially in high-sensitivity systems such as cold chains where data integrity directly impacts product safety. CryoTrack treats UX as a core enabler of this segment's participation, because low adoption among low-infra actors means incomplete data entering the ledger, which reduces the reliability of every checkpoint in the network and to achieve this, a **zero-friction, mobile-first design philosophy**, where all backend complexity is abstracted away from the end user, is adopted. The primary entry point for ground-level actors is a native mobile app. The decision to build a native app rather than a web-based solution was a deliberate technical one: OCR inference requires direct camera hardware access and reliable local processing that a browser environment cannot consistently guarantee on low-end devices. Since the OCR cross-check of physical displays against digital logs is a core integrity feature of Tier 1, compromising it for the sake of easier distribution was not an option. The app is the mechanism through which a small farmer, a local cold storage operator, or an independent transporter becomes a cryptographically verifiable participant in the same network as a large pharmaceutical distributor, with no specialized hardware and no enterprise software training required. At a checkpoint, the workflow is reduced to its bare essentials: scan the shipment QR code, capture the required physical proofs and documentation for that goods type, receive an immediate Tier 2 pre-transfer evaluation, and confirm or dispute the handoff with a single action. Every complexity of the Hyperledger Fabric backend, the deferred endorsement signing, and the AI evaluation is entirely invisible to the worker. The app is also built for India's connectivity reality, with Tier 1's offline caching ensuring that a worker inside a cold storage facility or a remote highway corridor can complete a checkpoint without network access, with all data anchored to the ledger automatically once connectivity is restored.

## 3. Blockchain Applicability

Supply chain management is one of the most powerful and natural use cases for blockchain technology. The ideas of immutable data, trustlessness and decentralisation are a perfect fit for a supply chain. The underlying trust architecture for India's massive, fragmented network fundamentally demands a decentralized ledger. Centralized databases can be better in certain scenarios but they come with limitations like single point of failure, trust assumption that the centralised database will act true, data leak vulnerabilities and adaptability friction. Blockchain technology addresses these problems because it replaces these vulnerable, centralized points with a distributed, "trustless" environment. It also brings immediate wider benefits to the supply

chain, ensures cryptographic security for all data transfers, and automates trust across competing parties. Beyond these advantages, this decentralized architecture serves as the foundational layer needed to enforce our game-theoretic Checkpoint Mechanism and tackle the specific, critical vulnerabilities of the Indian logistics sector:

### **3.1 Immutability: The Antidote to the GIGO Vulnerability**

In centralised systems, the GIGO (Garbage In, Garbage Out) vulnerability thrives because you can't verify that a manual or digital log was not altered after addition and leaves a blanket of trust and increased chances of GIGO and data manipulation, such as changing a temperature reading hours after a cold-chain failure to avoid financial penalties. Blockchain's immutability fundamentally eliminates this risk. Once an actor accepts goods at a checkpoint and assumes liability, that transaction is cryptographically hashed and permanently anchored to the Hyperledger Fabric ledger which can be seen by all the actors of the ledger. It becomes a Write-Once-Read-Many (WORM) record. It cannot be deleted, manipulated by a localized bribe, or altered by a malicious administrator, ensuring the historical state of the goods is undeniably secure.

### **3.2 Decentralization: Unifying a Fragmented Market**

Digitising India's large fragmented and untapped market seems only rational with Blockchain because with centralised approaches forcing crores of unorganized actors to integrate into a single, monopolistic enterprise ERP system is practically, economically impossible and also has a threat of single point of failures and attack points which can be catastrophic for supply chains. Blockchain's decentralized architecture allows these independent actors to participate in a unified peer-to-peer network. No single corporation owns the network; instead, the protocol itself governs the interactions. This decentralization democratizes supply chain security, allowing a small local transporter to securely interact with a massive pharmaceutical distributor on equal, cryptographically verifiable footing.

### **3.3 Transparency and Auditability: Instant Traceability**

For centralised and most of the current solutions tracing a compromised batch of goods, like the tragic cases of contaminated cough syrup in Ramnagar or The Gambia requires manually digging through the disconnected databases of multiple logistics providers. This process takes weeks, and in healthcare, those delays cost lives. Blockchain eliminates these silos by creating one shared, tamper-proof record. Every stakeholder, including regulators like the FSSAI, gets secure, role-based access to the same data. If a bad product reaches a pharmacy, auditors don't have to guess; they can instantly trace the product's journey backward. The ledger pinpoints exactly where the compromise happened and the evidence from the digital and manual logging can be used to decide who is liable, turning a weeks-long investigation into an instant search.

## 4. Technical Approach and Architecture

CryoTrack's architecture is built and designed with the goal to provide a solution to the entire Indian market from farmers to enterprises. It's a decoupled, microservices-based framework with pluggable, customisable and modular components. To serve the wide market spectrum the system must be lightweight at the edge but enterprise-grade at the core. We achieve this by separating the user interface, artificial intelligence engines, integration layers, and the blockchain ledger into distinct, interoperable modules.

### 4.1 Modular "Plug-and-Play" Architecture for Cross-Industry Scaling

For true scalability and satisfying India's diverse supply chain needs, CryoTrack uses a decoupled, "plug and play" modular architecture. Instead of building a single restricted framework, we are building an open source protocol where we have separated the underlying trust infrastructure from the industry specific data layers. This design philosophy ensures that scaling into a new sector is a standard. Like starting with pharmaceuticals, expanding to agriculture or adding support for a rare fruit which has extra supply chain steps in the future is seamless and fast because it does not require rebuilding the core application. Instead we simply built these modules that fit in directly into the existing CryoTrack framework. For example, adding support for a new supply chain such as dairy requires only defining the required checkpoint inputs (e.g., temperature, storage duration) and validation thresholds. Once defined, this module is dynamically loaded by the system, allowing all interfaces and AI engines to adapt automatically without requiring changes to the application logic.

The base layer which is the foundation of this system is the Universal core, a common layer across these modules the Hyperledger Fabric network, the Checkpoint mechanism, cryptographic identity management and the zero sum liability transfer logic. By design the trust and security layer expands to all supply chains with the ability to select different resolution entities, this fits right in the modular and secure design of CryoTrack. Because of the base layer we can seamlessly stack Dynamic Input modules on top of it. These input modules handle the physical handoff requirements, data logging and other steps in a supply chain. Throughout the WebApp, mobile app and API gateway, the system uses dynamic input rendering to adapt for the specific supply chain, meaning ground-level workers never have to download a new or separate application.

Furthermore, this modular design boosts the performance of our Tier 1 and Tier 2 AI by providing valuable context which differs for each supply chain. The AI frameworks will be designed to use this context and provide better results. For example the anomaly detection thresholds for a frozen vaccine are different from those of a fresh product. The supply chain context passed through the module allows the AI to immediately recognise which external APIs to call such as FSSAI compliance databases for food, CDSCO guidelines for pharma or custom supply chain guidelines if any. The system automatically adjusts its Integrity Trust Score calculations accordingly. By compartmentalizing the supply chain inputs and data analytics into these distinct, swappable modules, CryoTrack functions as a universally replicable protocol, enabling rapid deployment into any new vertical.



## **4.2 API-First Integration Support**

### **API Gateway and Key Management**

To maximise adoption, CryoTrack implements a decoupled, two-tier security architecture that strictly separates Web2 API access from Web3 cryptographic transaction signing. Using common protocols (API Keys, OAuth 2.0, or JWTs), external systems like old ERPs or the backend of mobile apps will first authenticate with the API Gateway. Before any computational load reaches the blockchain, this outer layer mitigates attack vectors, enforces rate limitation, and verifies payload schemas. Then the Web2 authentication is converted to a verifiable Web3 identity to verify cryptographic identity. Large-scale businesses need automated, high-throughput machine-to-machine communications, so a cloud-based Hardware Security Module (HSM) securely manages their Fabric private keys. Automated integrations are made possible by the HSM, which programmatically signs ledger transactions on the company's behalf without ever disclosing the private keys.

On the other hand, cryptographic key management is completely self-custodial and inbuilt in the edge device for MSME ground-level employees using the CryoTrack mobile app. The private key is permanently stored in the device's hardware-backed keystore (iOS Secure Enclave or Android Keystore), while the mobile app creates a local keypair during onboarding and registers the public key with the Fabric Certificate Authority (CA). Instead of depending on the centralized API Gateway, transactions like admitting liability during a handoff are cryptographically signed directly on the employee's smartphone. By demonstrating mathematically that the signature came from the employee's actual device, this localized signing architecture ensures complete non-repudiation. Importantly, it also makes Tier 1's offline caching features possible, enabling employees to safely sign and store verifiable cryptographic checkpoints in places with no connectivity (such as distant highways or cold storage units) until network access is restored.

### **Event-Driven Architecture and Asynchronous Processing**

The API Gateway uses an event-driven design to manage large amounts of telemetry and handoff data. High-frequency data inputs are digested via gRPC and sent to an internal message broker instead of depending on synchronous, blocking queries to the ledger. This significantly lessens network congestion by enabling the gateway to batch process telemetry changes before sending them to the Fabric ordering service. Additionally, the gateway listens for freshly committed blocks and chaincode events using Fabric's Channel Event Hubs. The gateway immediately initiates outbound webhooks at the completion of a liability transfer (checkpoint) on-chain, updating the business ERPs in real-time without the need for ongoing polling.

## **Secure Data Exposure for ESG and Scope 3 Auditing**

The API allows third-party developers and auditing firms to create specialized applications on top of the CryoTrack protocol, going beyond ordinary supply chain tracking. It does this by exposing safe, permissioned endpoints. Verifiable Environmental, Social, and Governance (ESG) compliance is one of its main uses. Companies have always been reluctant to provide secret supply chain routing to outside auditors, despite the fact that tracking Scope 3 carbon emissions is an essential organizational necessity. Fabric's Private Data Collections (PDCs) are used by CryoTrack's API to address this. Through GraphQL, the gateway may provide external ESG auditing methods with verifiable, timestamped transit metrics (such as precise route duration, vehicle telemetry, and idling periods). While the underlying proprietary data such as cargo pricing, precise goods volume, and sensitive trade relationships remains isolated and invisible to the auditing application, auditors receive mathematically provable Scope 3 carbon footprints because the data is cryptographically anchored to the ledger.

## **4.3 Core System Components**

### **User Interfaces: The Edge App and Management Dashboard**

CryoTrack uses a dual interface strategy to cater to the vastly different operational needs across the supply chain. The primary point of interaction for ground level workforce is the CryoTrack mobile app especially designed considering the conditions it will be mostly used in. For supply chain managers, enterprise auditors and other actors, CryoTrack provides a comprehensive Website dashboard. The robust interface offers deep macro level oversight, special node level actions, complex data analytics and dispute resolution tools, ensuring management teams stay perfectly in sync with edge operations.

### **The Intelligence Layer: Edge and Centralized AI**

Integrated with the interfaces is our two-tiered AI layer. The Tier 1 operates directly within the mobile app. Meanwhile, the Tier 2 central engine operates a cloud hosted model which handles the heavy lifting and provides data to both the interfaces.

### **API and Integration Gateway: The Architectural Middleware**

To facilitate communication between all the different components, external systems and modules, the API and integration gateway is the core middleware for CryoTrack. This component manages entire traffic routing, payload formatting and data transmission across

protocol. It acts as a critical bridge that allows enterprise legacy systems, such as existing SAP or Oracle ERPs, to interact directly with the decentralized network instead of running a node and getting raw data themselves.

## **Ledger and Consensus Engine: The Immutable Foundation**

The core base layer of CryoTrack is the Ledger and the Consensus engine. This component consists of the immutable ledger and a CouchDB which allows us to perform fast and complex queries of the supply chain's state. It is responsible for executing the custom Checkpoint chaincode, which enforces the zero sum liability transfer at every handoff. By employing our highly optimized Deferred Endorsement policy, this consensus engine rapidly validates the cryptographic signatures sent from the Mobile App and enterprise APIs. It sequences these validated transactions into Write-Once-Read-Many (WORM) blocks, creating a definitive, unalterable historical record of the asset's journey.

## **4.4 Interactions and Data Flow: The Checkpoint Lifecycle**

### **On-Chain Consensus and Strategic Update**

All cryptographically signed payloads are transmitted through the API gateway to the Hyperledger fabric network. Utilising our scalable deferred endorsement policy, the required peer instantly validates the signature, permission and the smart contract logic. The transaction is then sequenced in the RAFT nodes and committed to the ledger.

### **Asset Genesis: System Initiation**

For a complete supply chain, CryoTracks data lifecycle should preferably begin at the origin point, it also supports supply chain from any point, even a single actor. This initiation phase is where the asset is first digitized and gives a cryptographic identity in the ledger. The origin worker uses the mobile app, website or custom enterprise app with our API's integrated to log the new batch of goods. Because of our modular design, the worker selects the good type and required inputs are requested. Upon submission, the Hyperledger fabric smart contract executes, generating the initial digital twin of the asset and assigning 100% of the initial liability to the originating actor

### **The Handoff Checkpoint: Context Aggregation**

Once the asset is digitized, the recurring “Handoff” phase is the physical transfer of goods from one actor to another. The underlying Checkpoint mechanism remains the same for each good, the required data inputs seamlessly adapt to the specific good based on the supply chain using the specific module. The handoff initiates when the receiving actors opens their Mobile App and scans the shipment's QR code. The app securely calls the API Gateway, which queries the ledger for the current ownership state and a ledger state update is made in the goods using the receivers key and starting the verification time.

## **Edge Verification and Liability Transfer**

In the verification period, the Mobile App actively prompts the receiving actor to capture the physical proofs required for that specific goods like the condition of the cargo boxes etc. The Tier 1 Edge Engine processes the localized data directly on the worker's smartphone, extract any data that is needed for cross verification with the ledger and make a call to the central AI engine. The AI prompts the confidence score and hints steps for generating proof if anything is wrong. Now within the verification period if no dispute is made or the accept option is chosen, the liability is transferred to the receiver automatically by timestamp in case of no dispute or by overriding using the accept transaction.

## **5. Benchmarking and Differentiation**

CryoTrack operates at the intersection of three existing solution categories: enterprise blockchain supply chain platforms, India-specific agri and logistics technology, and cold chain IoT monitoring tools. The analysis below evaluates CryoTrack against four significant players across these categories. The intent is not to diminish existing solutions, many of which serve their intended markets well, but to identify the specific gaps in the Indian logistics context that motivated CryoTrack's design decisions.

### **5.1 Competitor Analysis**

#### **5.1.1 IBM Food Trust (Hyperledger Fabric)**

IBM Food Trust is the most technically comparable platform, also built on Hyperledger Fabric and targeting food supply chain traceability, with Walmart, Nestle and Dole among its adopters. Its deployment at Walmart reduced trace time for a food item from seven days to 2.2 seconds, representing the most cited benchmark for what the Hyperledger Fabric foundation can achieve at enterprise scale. It is designed for large, already-digitised enterprise supply chains where onboarding requires significant IT infrastructure and IBM cloud dependency, making it economically and technically inaccessible to India's MSME-driven logistics sector. It has no India-specific regulatory integration with systems such as ULIP, VAHAN, e-Way Bill or FSSAI, and no mechanism to verify or challenge data inputs at the ground level. CryoTrack shares the same Hyperledger Fabric foundation but is oriented toward a different problem: creating

trustworthy data in environments where the infrastructure and incentives for honest reporting do not naturally exist.

### 5.1.2 TraceX Technologies (India)

TraceX is an Indian blockchain-based agri traceability platform focused on farmer-to-fork transparency, ESG reporting, sustainability certification and carbon credit tracking. Its adoption model targets cooperatives and organised agri exporters rather than the fragmented MSME transporter-warehouse-kirana network. It records what actors report without a mechanism to verify or challenge those inputs at the ground level. It does not cover real-time cold chain integrity or in-transit condition monitoring. CryoTrack and TraceX address meaningfully different segments of the agri supply chain, with TraceX oriented toward certification and export compliance and CryoTrack oriented toward domestic transit integrity and safety.

### 5.1.3 Arya.ag (India)

Arya.ag is a leading Indian agri-fintech platform focused on post-harvest commodity management, warehouse receipt financing and storage optimisation. It is primarily a financing and storage management platform rather than a supply chain traceability or cold chain monitoring tool, and does not cover the transit leg of the supply chain, which is the most vulnerable phase for cold chain failure and data falsification. CryoTrack and Arya.ag are largely complementary rather than competing, addressing different layers of the same supply chain.

### 5.1.4 Eka Platform (Commodity and Supply Chain ERP)

Eka is an enterprise-grade commodity management and supply chain platform used by large trading houses and processors across agri, energy and metals. It offers mature enterprise analytics and deep procurement and contract management workflows. It is not blockchain-native, relying on a centralised SaaS architecture, and is focused on trading workflows rather than physical transit integrity or cold chain condition verification. It has no ground-level mobile interface or pathway to MSME adoption.

## 5.2 Comparison Matrix:

| Dimension     | IBM Food Trust | TraceX | Arya.ag | Eka Platform | CryoTrack |
|---------------|----------------|--------|---------|--------------|-----------|
| TARGET MARKET |                |        |         |              |           |

|                                 |                                                    |                          |                                    |                                |                                                                                                                  |
|---------------------------------|----------------------------------------------------|--------------------------|------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Primary sector</b>           | Large enterprise food chains                       | Organised agri exporters | Commodity traders and agri-fintech | Large commodity trading houses | MSME-to-enterprise across pharma, agri and food logistics                                                        |
| <b>MSME accessibility</b>       | Not designed for                                   | Organised agri only      | Commodity traders only             | Not designed for               | Core design target; smartphone-only entry point                                                                  |
| <b>ARCHITECTURE</b>             |                                                    |                          |                                    |                                |                                                                                                                  |
| <b>Blockchain foundation</b>    | Hyperledger Fabric                                 | Blockchain-based         | Centralised database               | Centralised SaaS               | Hyperledger Fabric                                                                                               |
| <b>Consensus and throughput</b> | Standard multi-org endorsement; ~140 TPS published | Not published            | Not applicable                     | Not applicable                 | Deferred single-signature endorsement; targets 3x to 5x over multi-org baseline                                  |
| <b>Single point of failure</b>  | Low                                                | Low                      | High                               | High                           | Low                                                                                                              |
| <b>API integration</b>          | Enterprise only                                    | Partial                  | Partial                            | Enterprise only                | REST, GraphQL and gRPC via API gateway alongside mobile app                                                      |
| <b>DATA ENTRY AND INTEGRITY</b> |                                                    |                          |                                    |                                |                                                                                                                  |
| <b>Primary data entry</b>       | Enterprise system integration                      | Manual actor reporting   | Manual actor reporting             | Enterprise system integration  | API integration for infra-heavy actors; OCR document digitisation, guided structured input and timestamped photo |

|                                       |                            |                            |                            |                            |                                                                                                                                  |
|---------------------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                                       |                            |                            |                            |                            | logging for low-infra actors                                                                                                     |
| <b>Ground-level data verification</b> | None; assumes honest input | None; assumes honest input | None; assumes honest input | None; assumes honest input | Structured capture replacing manual free-text; OCR digitisation of paper documents; GPS-tagged photo evidence at each checkpoint |
| <b>Offline capability</b>             | No                         | No                         | No                         | No                         | Local caching with automatic ledger anchoring on reconnect                                                                       |
| <b>IN-TRANSIT MONITORING</b>          |                            |                            |                            |                            |                                                                                                                                  |
| <b>Real-time condition monitoring</b> | Basic                      | No                         | No                         | No                         | AI-assisted telemetry evaluation at each liability transfer                                                                      |
| <b>External signal correlation</b>    | No                         | No                         | No                         | No                         | Cross-references telemetry against weather, route and traffic data before acceptance                                             |
| <b>Pre-transfer evaluation</b>        | No                         | No                         | No                         | Partial post-hoc analytics | Tier 2 evaluation fires at moment of liability transfer; actor retains full decision autonomy                                    |
| <b>Explainable AI outputs</b>         | No                         | No                         | No                         | No                         | Every flag surfaced as plain-language explanation with specific triggering signals                                               |

| TRACEABILITY                              |                              |                         |                         |                         |                                                                                                                                   |
|-------------------------------------------|------------------------------|-------------------------|-------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Audit trail</b>                        | Immutable on Fabric          | Immutable on blockchain | No                      | No                      | Immutable WORM-enforced on Fabric ledger                                                                                          |
| <b>Trace time per incident</b>            | 2.2 seconds at Walmart scale | Not published           | Not applicable          | Days to weeks manually  | Targets comparable structured query time (< 5 seconds) compared to current baseline estimated at 2 weeks across fragmented actors |
| DISPUTE HANDLING                          |                              |                         |                         |                         |                                                                                                                                   |
| <b>Resolution mechanism</b>               | No structured mechanism      | No structured mechanism | No structured mechanism | No structured mechanism | Permissioned auditor nodes with access to structured tamper-proof on-chain evidence per handoff                                   |
| REGULATORY AND COMPLIANCE                 |                              |                         |                         |                         |                                                                                                                                   |
| <b>India government API integration</b>   | No                           | Partial                 | Partial                 | No                      | ULIP, VAHAN, FASTag and e-Way Bill as verification oracles                                                                        |
| <b>Sector regulator alignment</b>         | No                           | FSSAI partial           | APEDA partial           | No                      | Verified Compliance Records for FSSAI, CDSCO and APEDA                                                                            |
| <b>National Blockchain Framework 2024</b> | No                           | Partial                 | No                      | No                      | Aligned                                                                                                                           |



|                      |              |         |         |         |                                        |
|----------------------|--------------|---------|---------|---------|----------------------------------------|
| <b>DPDP Act 2023</b> | GDPR-focused | Partial | Partial | Partial | Data minimization with no PII on-chain |
|----------------------|--------------|---------|---------|---------|----------------------------------------|

### 5.3 Core Differentiation

The common characteristic across all reviewed solutions is a shared foundational assumption: that the data being recorded into the system is honest. Existing platforms build traceability, analytics and compliance reporting on top of inputs that actors are expected to enter truthfully, without a structural mechanism to verify or challenge those inputs at the ground level. This is a reasonable assumption for large, organised, already-digitised enterprise supply chains. It is not a reasonable assumption for India's fragmented logistics sector, where manual log falsification, cheap unencrypted sensors and financial pressure to conceal cold chain failures create a pervasive GIGO vulnerability at the ground level. CryoTrack's design starts from this structural reality rather than assuming it away. The Checkpoint Mechanism uses game-theoretic incentive design to make honest data input the rational choice for every actor in the chain. The on-device OCR verification addresses a specific and documented manipulation tactic rather than a theoretical one. The Tier 2 real-time evaluation gives the accepting actor the information they need to make an informed liability decision rather than relying on trust in the previous actor. These are not incremental improvements on existing approaches but responses to a different and harder problem than existing solutions were designed to solve.

## 6. Non-Crypto Blockchain Platform

### 6.1 Selected Platform: Hyperledger Fabric

We selected Hyperledger Fabric as our foundational non crypto blockchain platform which is the base and core layer of Cryotrack. It is an enterprise-grade, permissioned distributed ledger technology (DLT) and designed for B2B solutions where identity, privacy and high throughput performance are needed with the core benefits of blockchain. Fabric provides abilities for a custom configuration using which we employ a RAFT consensus mechanism which is lighter and more secure based on our Deferred Endorsement Policy and Checkpoint Mechanism.

### 6.2 How Hyperledger Fabric Addresses the Identified Problems

Permissionless blockchains can be slow and expensive depending on user presence which can be real issues when building a solution for scalability, speed and security. The structural requirements that we identified for a supply chain solution, to address GIGO, the economic constraints of MSMEs and the regulatory compliance needed, we needed a core layer which can be used to address all of them. Hyperledger Fabric eliminates the need for gas fees and tokenomics addressing the MSME onboarding issues. Due to this, CryoTrack offers a flat, predictable cost allowing actors to interact with the ledger thousands of times a day with marginal cost per transaction.

Beyond economics, the Handoff Liability model will not work on public blockchains where we need economic certainty which becomes a real issue for onboarding supply chain actors. Fabric's permissioned nature fundamentally solves these by allowing KYCed and approved users. With Fabric we will utilize a robust Membership Service Provider (MSP) to manage X.509 certificates, ensuring every participant is a known verified entity. Furthermore, privacy is built into Fabric and often lost in public blockchains. CryoTrack protects sensitive commercial interests involving pricing, routing and counterpart agreements through Channels and Private Data Collections supported by Fabric. Using this we can isolate transactions so that competing logistics providers can exist in the same network while keeping their data secure as often enterprises don't adapt a system where competitors can view their transaction histories.

Finally, Hyperledger Fabric satisfies the needs of CryoTrack handling supply chains across India by delivering high throughput and flexible integration, enhanced further by our custom consensus. Public chains rely on anonymous miners and big pool entities competing or in line for different mining algorithms such as Proof of Work or Proof of Stake. These mechanisms are prone to high latency, ghost blocks, and censorship. While Fabric uses deterministic consensus, will allow a transaction to final instantly which is a critical requirement for supply chains. Combined with our custom Deferred Endorsement Policy, Fabric can efficiently process thousands of writes and updates per second. Additionally, Fabric allows smart contracts to be written in standard programming like Javascript which allows us to build our decoupled, API first architecture. This flexibility enables seamless integration with our Two-Tier AI system and legacy enterprise ERPs without introducing severe technical friction.

## **7. Permissioned Blockchain and Smart Contracts**

### **7.1 Strategic Deployment of a Permissioned Ecosystem**

The true value of CryoTrack lies in how we structure the permissioned ecosystem. Operating on a consortium model, fabric ensures verified and authorised actors can interact with the ledger. To onboard India's unorganised logistics, we will have a streamlined onboarding and KYC mechanism specifically designed for MSMEs. A local transporter or kirana owner can directly register through CryoTrack by providing the relevant details and operational credentials which are then routed to designated permissioned nodes for approval. After approvals they can participate in handoffs, be a part of the ledger where they can be identified by enterprises etc.

The permissioned architecture also enables sophisticated data segregation through multi-channel architectures. Logistic providers often provide service to competing brands,

commercial confidentiality is crucial. By using the specialized channels, data between two actors like a pharmaceutical manufacturer and its contracted distributor can share a secure, isolated ledger for specific shipments. This enhances security by ensuring data is only disseminated on a strict need-to-know basis, while still allowing government bodies like the CDSCO or FSSAI to operate overarching auditor nodes with read-only access for immediate, network-wide compliance verification.

## **7.2 Chaincode-Driven Business Logic**

To implement the Handoff liability model to life, CryoTrack uses the highly customized smart contracts, known within Hyperledger Fabric as Chaincode. These contracts are the autonomous enforcers of the supply chain rules from required fields to accessibility enforcement. When a new batch of goods is digitized at the origin point, the specific Service Level Agreements (SLAs) such as temperature thresholds for a vaccine, maximum allowed transit times, and the mandatory proof requirements are built into the localized smart contract instance and enforced before any transaction is added. The chaincode intercepts any payload, validating the digital signatures and deterministic proofs against the immutable on-chain SLAs. It acts as an uncompromising digital referee, verifying the transaction's validity before allowing the ledger state to update.

## **7.3 Enhancing Functionality and Autonomous Security**

The security of supply chain compounds drastically with the fusion of permissioned access, off chain intelligence through our two Tier AI models and the autonomous chaincode implementing the Checkpoint mechanism. Functionally, it creates a seamless, automated dispute management system. The Tier 2 AI engine detects anomalies and presents a full traceable map of the supply chain. This eliminates manual auditing delays and reduces likelihood of a compromised asset from moving further down the chain under false pretenses. For a handoff or any update the data is cryptographically signed and executed across the network validating the endorsement policy, the rules of engagement are tamper proof. No single entity like a powerful enterprise, a localized malicious actor, or even CryoTrack's internal administrators can bypass the chaincode to alter the SLAs or force a clean liability transfer for a compromised shipment. This guarantees that compliance is mathematically enforced by the protocol itself, transforming a fragmented, low-trust physical network into a highly secure, autonomous digital ecosystem.

## **8. Regulatory Framework**

CryoTrack is designed to operate within India's regulatory ecosystem for digital logistics infrastructure, supply chain traceability, and data governance. The platform aligns with national legislation governing electronic records, logistics digitization, and personal data protection while enabling regulated industries to maintain verifiable compliance records across their supply

chains.

| Regulation/Framework                      | Compliance Requirement                                                                                          | Cryotrack Alignment                                                                                                                                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Information Technology Act 2000           | Legal recognition of electronic records and admissibility of digital evidence.                                  | Cryptographically timestamped records create immutable audit trails that support compliance verification and regulatory investigations.                                                                                                                                           |
| Digital Personal Data Protection Act 2023 | Protection and responsible processing of personal data in digital systems.                                      | CryoTrack follows data minimization principles and records operational events without storing unnecessary personally identifiable information.                                                                                                                                    |
| National Logistics Policy 2022            | Promotion of digital integration across logistics networks to improve supply chain efficiency and transparency. | CryoTrack is designed to integrate with the Unified Logistics Interface Platform (ULIP) to access logistics APIs, including shipment documentation verification through e-Way Bill systems, vehicle registry validation via VAHAN, and route verification using FASTag toll data. |

|                                                    |                                                                                           |                                                                                                                                                                                                    |
|----------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| National Blockchain Framework 2024                 | Standardization of enterprise blockchain infrastructure for secure digital recordkeeping. | CryoTrack is built on <b>Hyperledger Fabric</b> , a permissioned enterprise blockchain aligned with interoperability and security standards promoted under the National Blockchain Framework 2024. |
| Goods and Services Tax Act 2017 (e-Way Bill Rules) | Mandatory documentation and verification of goods transportation.                         | Shipment events recorded on CryoTrack can be validated against official logistics documentation systems.                                                                                           |

## 9. Security, Privacy, Scalability and Performance

### 9.1 Security and Zero-Friction Embedded Wallets

We at CryoTrack prioritize security at most and want to have as minimal security dependencies as possible. We shift the burden of trust from manual oversight to cryptographic certainty, ensuring non repudiation at every node of the supply chain. Key management is a major security challenge especially in the case of MSMEs. We can't expect ground level workers to manage and store keys. To solve this without compromising security, CryoTrack utilizes self custodial, embedded wallets directly within the edge mobile application. When a new actor registers, we automatically generate a cryptographic key pair locally on the device, storing the private key inside the hardware backed secure enclave or encrypted format.

The goal for this approach is to remove the trust needed for actors to trust CryoTrack for key management. While this edge bound security prevents remote compromise, it needs a robust recovery process like device loss or any reason in which the key may be lost. Because CryoTrack works on a permissioned ledger and not a public chain, a lost key does not mean lost identity. If an actor's key is lost, they can initiate a secure account recovery protocol where the actor goes through a streamlined re-KYC process to prove their real world identity. Upon verification, the network's Membership Service Provider (MSP) explicitly revokes the compromised device's cryptographic certificate, neutralizing any risk of malicious use.

Simultaneously, a new key pair is generated on the worker's replacement device and mathematically bound to their existing identity profile on the ledger. This recovery mechanism keeps the historical Verified Compliance Record intact while securely restoring their ability to participate in the supply chain.

Based on this method and design, the wallet interface is entirely abstracted away from the user, providing an easy and traditional experience with Web3 security. The crucial point being that even CryoTracks servers never have access to the key emphasizes on the security considerations as this guarantees that every liability transfer is cryptographically bound to the specific verified actor's device, mathematically preventing impersonation, unauthorized ledger updates, or central database manipulation by internal administrators.

## **9.2 Privacy and Commercial Confidentiality**

We have already discussed how data privacy is not just a regulatory requirement but a commercial necessity and how we use Channels and Private Data collections to address them. Basically giving the ability for competitors to operate on the same overarching CryoTrack protocol while remaining completely blind to each other's operational data. To strictly adhere to the Digital Personal Data Protection (DPDP) Act 2023, we employ aggressive data minimization using specific designed user actions and Tier 1 AI. No personally Identifiable Information (PII) is ever stored directly on the immutable ledger. We only use cryptographic hashes, aggregated telemetry data and compliance flags to anchor data on chain ensuring data privacy while maintaining data authenticity and maintaining a verifiable audit trail for regulators.

## **9.3 Scalability and High-Throughput Performance**

Both scalability and high throughput performance are faces on the same coin and are crucial for a solution aiming for enterprise and large Indian markets. We target scalability by an agnostic and modular architecture supported by our core layer, Hyperledger Fabric. Fabric allows us to onboard new actors and new goods with minimal cost. This permissioned framework allows us to achieve massive economic scalability, allowing ground-level actors to perform thousands of handoffs and data updates at zero marginal cost per transaction.

Technically, performance is boosted by the integration of the deterministic RAFT consensus mechanism and our custom Deferred Endorsement Policy. By reducing the heavy multi-signature requirements typically bottlenecking standard Fabric deployments, we effectively bypass standard endorsement latency. This allows the system to scale rapidly, increasing the number of transactions and telemetry verifications process per second when compared to traditional solutions.

## 9.4 System Robustness and Reliability

A supply chain solution is only robust when it has the ability to survive real world operational hazards which are more common in India than most of the world. We guarantee reliability through a decentralized, single point of failure resistant architecture at the core, paired with resilient edge computing at the ground level.

Our Tier 1 AI and mobile application are built with robust offline caching capabilities considering the scenarios often the actors work in like a cold storage facility, even with no network the application will work for the crucial steps. This allows the worker to sign handoffs, locally hashes the proofs, signs and caches it. The moment connection is restored the application automatically anchors the batch to the ledger, preserving the exact timestamp and integrity of the timeline. This ensures 100% operational uptime and reliable data continuity regardless of localized infrastructure failures.

## 10. Execution Plan

CryoTrack will be rolled out in phases, starting from points in the supply chain where control naturally exists, such as warehouses, cold storages, and aggregation centres. These are places where goods are received, checked, stored, and dispatched, making them the most practical starting point for introducing traceability.

Instead of trying to do a large-scale roll out from the starting, we will follow a **build => test => refine => scale** approach to ensure reliability in real-world Indian conditions.

### 10.1 Core Product Development ( Phase 1 )

In the first phase, the focus will be on building a core product that people can actually use in real operations. This includes developing a mobile-first application for ground-level actors like warehouse operators, handlers, and supervisors. At the same time, basic checkpoint workflows such as goods arrival, verification, storage logging, and dispatch will be implemented. The on-device integrity layer will also be introduced, where the system performs simple checks (like verifying whether values are within allowed limits), supports OCR-based verification, and works even in low or no connectivity through offline data storage. On the backend, the Hyperledger Fabric network will be set up with basic chaincode to handle asset creation, checkpoint logging, and liability transfer. An initial API gateway will also be developed to allow smooth communication between different parts of the system.

#### Key milestone:

A fully functional MVP where a shipment can be created, transferred, and verified across checkpoints.

## **10.2 Advanced Logic and Integration Layer ( Phase 2 )**

Once the base system is stable, the second phase will focus on adding deeper intelligence and stronger validation layers. A centralized AI engine will be deployed to detect anomalies and assist in decision-making during checkpoints. The system will also integrate with government infrastructure through ULIP APIs to enable external verification where required. Dispute creation and resolution workflows will be introduced so that issues can be formally raised and resolved within the system. Alongside this, a management dashboard will be built for enterprise users and auditors to monitor operations. Security will also be strengthened through proper identity and access management, role-based permissions, and controlled data sharing.

### **Key milestone:**

A production-ready system with intelligence, dispute handling, and monitoring capabilities.

## **10.3 Pilot Deployment and Field Testing ( Phase 3 )**

In the third phase, CryoTrack will be tested in a real-world pilot setup within a specific supply chain corridor, such as a cold storage to transporter to distributor flow. A small group of warehouses, transporters, and distributors will be onboarded to test the system under actual working conditions. The goal here is to observe how checkpoint workflows perform in daily operations, especially in environments with limited connectivity. This phase will also help identify how users interact with the system and what improvements are needed to make it easier for non-technical users. Success will be measured by improvements such as fewer manual logging errors, better compliance at checkpoints, faster dispute resolution, and overall system reliability in real conditions.

### **Key milestone:**

Successful completion of live shipments using CryoTrack with measurable improvements in traceability, data quality, and dispute handling.

## **10.4 Expansion Across Supply Chain Layers ( Phase 4 )**

After a successful pilot, the fourth phase will focus on expanding CryoTrack across different parts of the supply chain. This includes moving upstream to farmers, producers, and aggregation centres, and downstream to distributors, retailers, and kirana stores. The system will also be extended across multiple industries such as pharmaceuticals, agriculture, and food logistics. To support this growth, modular templates will be created for different supply chains so that onboarding becomes faster and easier. Tools will be developed to help MSMEs adopt the system without heavy setup, and APIs will allow integration with existing enterprise systems like ERP or warehouse management software. Additional analytics and compliance reporting features will also be introduced.



**Key milestone:**

Multi-actor, multi-layer deployment across at least one full supply chain.

**10.5 Multi-Region Scaling and Ecosystem Growth ( Phase 5 )**

In the final phase, the focus will shift to scaling across multiple regions and building a larger ecosystem. The rollout will happen state by state, starting with major logistics hubs and industrial clusters. Partnerships will be built with logistics operators, cold chain networks, and industry bodies to drive adoption. As more government datasets become available through ULIP, the system will integrate with them to improve verification capabilities. Over time, this will lead to the creation of a unified, interoperable, and verifiable supply chain network, positioning CryoTrack as a foundational trust layer for logistics in India.

**Key milestone:**

Cross-region deployment with repeatable onboarding model.

**10.7 Team and Resource Allocation**

The initial team will operate as a lean, cross-functional unit, where one developer will focus on backend and blockchain responsibilities including setting up Hyperledger Fabric, developing chaincode, and building APIs, another developer will handle the frontend and mobile side by building the mobile-first application and dashboard, while one person will manage product and operations, taking care of user flows, pilot coordination, and onboarding.

During the pilot phase, additional support will be required for **on-ground onboarding, training, and feedback collection**.

**10.8 Cost and Funding Approach (High-Level)**

The initial phases (Phase 1–3) will be executed in a cost-efficient manner by relying on open-source infrastructure such as Hyperledger Fabric, using cloud-based services for scalability, and operating with a small, focused team. Early-stage costs will primarily include cloud infrastructure, development efforts, and expenses related to pilot onboarding and operations. As the system moves beyond the pilot stage into Phase 4 and Phase 5, scaling will be supported through a combination of enterprise partnerships, government collaborations, and participation in grant programs.

**10.9 Development, Testing, and Deployment Approach**

For development and deployment, CryoTrack will follow a modular and iterative approach. Different components of the system will be built in parallel and connected through a common API layer. Testing will happen both in controlled environments and through real-world pilots to ensure the system works in practical conditions. Deployment will follow a phased rollout, starting small and gradually expanding across regions and supply chain layers as the system proves its reliability.

## **11. Expected Outcomes and Pilot Deployment**

### **11.1 Expected Outcomes**

CryoTrack is designed to fundamentally improve how supply chain data is created, verified, and used by introducing structured checkpoints and clear responsibility at every handoff. Instead of relying on fragmented records maintained by different actors, the system creates a consistent, verifiable flow of information that all participants can trust.

One of the primary outcomes is significantly improved traceability. Each movement of goods is recorded as a checkpoint, allowing stakeholders to track the journey of a shipment with minimal delay, even in low-connectivity environments. This reduces dependency on manual follow-ups and disconnected logs when investigating issues.

The system is also expected to reduce the impact of the “Garbage In, Garbage Out” (GIGO) problem, which is common in existing supply chains. By guiding users through structured workflows, enforcing basic validations, and requiring verification at handoff points, CryoTrack makes it significantly harder for incorrect or unchecked data to enter the system.

Another key outcome is clear and enforceable accountability. Every transfer of goods is tied to a specific actor, with responsibility explicitly recorded at each checkpoint. This removes ambiguity and allows stakeholders to pinpoint exactly where an issue occurred, rather than relying on assumptions across multiple parties.

CryoTrack also enables faster and more reliable dispute resolution. Since all events are time-stamped and backed by supporting data (logs, documents, or proofs), issues can be identified earlier and resolved based on evidence rather than manual reconciliation. This reduces operational delays and conflict between participants.

In addition, the system improves audit readiness. Since records are automatically generated during normal operations, businesses are better prepared for inspections and compliance checks without requiring separate documentation efforts. This creates a continuously updated, structured audit trail.

A critical long-term outcome is the inclusion of previously non-digitised participants into a formal supply chain network. By using a mobile-first approach that works on low-cost devices without requiring complex infrastructure, CryoTrack enables small warehouses, transporters, and kirana-level actors to participate in a structured and verifiable digital system for the first time.

### **11.2 Pilot Deployment Plan**

The pilot will be conducted in a controlled, real-world supply chain environment to validate CryoTrack under actual operating conditions. The goal is to test not just the technology, but also user behavior, adoption patterns, and system reliability in practical scenarios.

The scope of the pilot will be limited to a single supply chain corridor, for example a flow between a cold storage facility, a transporter, and a distributor. This ensures the system can be tested end-to-end while keeping the environment manageable.

A small but representative group of participants will be onboarded, including approximately two to three warehouses or cold storages, three to five transporters, and one to two distributors. This setup provides enough variation in operations to test real-world conditions without introducing unnecessary complexity.

The overall timeline includes both development and pilot execution. The first 8 to 10 weeks will be used to build and stabilize a focused, pilot-ready version of CryoTrack with limited but critical features. This will be followed by a 6 to 8 week pilot phase.

The pilot phase will begin with one to two weeks of onboarding and basic training, where participants are introduced to the system and initial configurations are completed. This will be followed by three to four weeks of live usage, where actual shipment flows are tracked using CryoTrack in day-to-day operations. The final one to two weeks will focus on monitoring system performance, collecting user feedback, and refining workflows based on real usage patterns.

### **11.3 Evaluation Metrics**

The success of the pilot will be evaluated using practical, outcome-driven metrics that reflect both system performance and real-world usability.

Checkpoint compliance rate will measure the percentage of handoffs completed using proper verification workflows, indicating how consistently the system is being used. Reduction in manual logging errors will be assessed by comparing data accuracy before and after CryoTrack adoption.

Dispute resolution efficiency will be measured by the time taken to identify and resolve issues, along with the quality of supporting data available during disputes. Early issue detection will track how many problems are identified at checkpoint stages rather than being discovered later in the supply chain.

Checkpoint dispute rate will be monitored to understand how frequently discrepancies are raised, providing insight into both system effectiveness and behavioral changes among participants.

User adoption and usability will be evaluated based on how consistently ground-level workers use the system and how easily they are able to complete required workflows without external support.

System reliability will be measured through successful data capture and synchronization, especially in low-connectivity environments, ensuring that no critical data is lost during operations.

## 11.4 Post-Pilot Outcome

Based on pilot results, CryoTrack will refine its workflows to better match real-world operational behavior, simplify the user experience for non-technical participants, and strengthen system reliability under varying conditions.

Once validated, the system will be expanded to additional participants, supply chain corridors, and industries. The learnings from the pilot will directly inform scaling strategy, ensuring that growth is grounded in proven usage rather than assumptions.

Over time, this approach will enable CryoTrack to evolve from a pilot solution into a scalable, network-driven infrastructure for building trust, accountability, and traceability across India's supply chains.

## 12. Government Department Collaboration

CryoTrack works alongside existing government systems rather than creating something separate. We use the data and infrastructure that the government has already built, and in return, government bodies get better visibility into supply chains they currently struggle to monitor.

We have identified five specific government bodies to work with. Two of them we can connect with right now without needing any special approval. The other three require a formal conversation and proposal, which we plan to initiate during Phase 2.

### Department of Promotion of Industry and Internal Trade (DPIIT) — ULIP

The government already has a system called ULIP (Unified Logistics Interface Platform) that connects data from 11 ministries, including e-Way Bill records, vehicle registration data from VAHAN, and FASTag toll records. Over 1,300 companies already use this.

CryoTrack uses ULIP as a live fact-checker at every handoff. When a transporter hands goods to a distributor, CryoTrack automatically calls ULIP to check: Is the e-Way Bill valid? Is this the same vehicle that was declared? Does the route match? If anything doesn't match, it gets flagged and recorded on the ledger before the receiving actor accepts responsibility.

We will register on [goulip.in](https://goulip.in), sign an NDA with NLDS, and get API access. This can be done in Phase 1 without waiting for any government approval.

**Benefit to DPIIT:** CryoTrack shows that ULIP data, when tied to a blockchain, can hold MSME-level logistics actors accountable, which is a core goal of the National Logistics Policy 2022.

## Ministry of Electronics and Information Technology (MeitY) — National Blockchain Framework

MeitY launched the National Blockchain Framework (NBF) in September 2024 with a budget of ₹64.76 crore. As part of this, NIC (National Informatics Centre) runs a Centre of Excellence in Blockchain that offers direct support to startups building on Hyperledger Fabric, which is exactly what CryoTrack is built on. NBF also has a ready-to-use sandbox called NBFLite, and a Smart Contract Library that already includes supply chain and drug track-and-trace templates.

We will register on [blockchain.meity.gov.in](https://blockchain.meity.gov.in), apply for NIC CoE support, and use NBFLite to test the system before the pilot. We will build on top of NBF's existing smart contract templates rather than starting from scratch. This gives us formal government support and makes our system easier to audit.

**Benefit to MeitY:** CryoTrack becomes a working example of NBF's supply chain template in action, something MeitY can point to as a real-world deployment across India's MSME logistics sector.

## APEDA — BHARATI Startup Accelerator

APEDA (Agricultural and Processed Food Products Export Development Authority) launched an accelerator called BHARATI in September 2025. It explicitly looks for startups working on blockchain traceability and cold chain monitoring which is exactly what CryoTrack does. Selected startups get three months of support including regulatory guidance, market access help, and exposure at international trade events like Gulfood and BIOFACH.

APEDA also runs Veg.Net, a traceability portal for agricultural exports. However, it does not currently cover what happens during transit, which is exactly the gap CryoTrack fills. We plan to apply to the 2026 BHARATI cohort and propose a live pilot on a specific crop corridor, such as grapes from Nashik or peanuts from Saurashtra both are APEDA-priority crops with well-documented transit losses.

A BHARATI selection gives CryoTrack a formal letter from the Ministry of Commerce, which is the strongest possible evidence of government collaboration.

**Benefit to APEDA:** CryoTrack gives APEDA a tamper-proof record of what happens to export goods between the farm and the port, a gap in their current system that they have acknowledged.

## FSSAI — Standards Integration and Audit Access (*Phase 2 — formal engagement required*)

FSSAI has clear standards for how food products must be stored and transported, specific temperature ranges for dairy, frozen foods, fresh produce, and meat. CryoTrack takes these published standards and embeds them directly into the chaincode as the rules that every food

shipment must comply with. There is no manual interpretation, the standards become part of the code.

We then give FSSAI a read-only auditor node on the CryoTrack network. During an inspection or a food safety incident, FSSAI can instantly pull up the complete custody and temperature history for any batch, instead of spending weeks collecting paper logs from multiple parties. This is directly relevant to tragedies like the Ramnagar poisoning and the Gambia syrup case, where the inability to trace products quickly cost lives.

We will engage FSSAI through their FoSCoS partnership program in Phase 2, with a proposal to formally pilot the audit node during the Phase 3 cold chain pilot.

**Benefit to FSSAI:** Real-time, tamper-proof cold chain visibility across thousands of MSME operators, something their current manual inspection model cannot achieve at scale.

Ministry of Food Processing Industries (MoFPI) — Cold Chain Scheme Integration  
*(Phase 2 — formal engagement required)*

MoFPI's Integrated Cold Chain scheme gives 35–50% capital subsidies to companies, cooperatives, and farmer producer organisations building cold storages and reefer van fleets. These are exactly the actors CryoTrack needs as its first set of verified ledger nodes.

We propose to MoFPI that CryoTrack be recommended as the digital compliance monitoring tool for all cold storage facilities that receive MoFPI subsidies. Every government-funded cold storage would come with CryoTrack built in, meaning the subsidy program automatically builds the network. NCCD's (National Centre for Cold Chain Development) published cold chain standards become the auto-generated SLA parameters for each cold storage module.

**Benefit to MoFPI:** Subsidy recipients can demonstrate compliance digitally rather than through manual paperwork, reducing post-investment monitoring burden on the ministry and on state food processing departments.

## **Business Model and plan for a Wider Proliferation Strategy**

CryoTrack's business model is built on "Trust-as-a-service" architecture. In the fragmented and low-trust Indian supply chain market, the model addresses the core economic friction. The "Data Producer" often has no incentive to provide honest data to the "Data Consumer" (the brand). Our strategy solves this by using an **Anchor-Payer model**, where value is captured at the enterprise level while ensuring zero-friction participation at the ground level.

Pricing Model

| Subcategory         | Individual<br>(Free)       | Essentials<br>(SME)<br><br>₹1,499 / mo | Premium<br>(Mid-Market)<br><br>₹7,499 / mo | Enterprise<br>(Corp)<br><br>Custom<br>Quotation |
|---------------------|----------------------------|----------------------------------------|--------------------------------------------|-------------------------------------------------|
| Asset<br>Management | 1 Location / 20<br>Events  | 5 Locations /<br>500 Events            | 25 Locations /<br>5,000 Events             | Unlimited                                       |
|                     | Standard<br>Trade Items    | Custom Trade<br>Items                  | Bulk Item<br>Management                    | Global SKU<br>Mapping                           |
| Data & Integrity    | 90-Day<br>Retention        | 1-Year<br>Retention                    | 3-Year Retention                           | Custom Data<br>Retention                        |
|                     | Handoff<br>Liability Logic | Mobile OCR<br>Data Capture             | Dispute<br>Verification Tools              | Tier-2 AI<br>(Reality Check)                    |

|                                 |                        |                            |                             |                                  |
|---------------------------------|------------------------|----------------------------|-----------------------------|----------------------------------|
|                                 | Standard Ledger        | Standard Ledger            | High-Throughput Ledger      | <b>Private Data Collections</b>  |
| <b>Connectivity</b>             | Mobile App Access      | Mobile + Web Dashboard     | <b>WhatsApp Bot Support</b> | <b>Full API / ERP Access</b>     |
|                                 | Zero-Friction Wallet   | Basic Exporting (CSV)      | API Webhooks (Basic)        | <b>Gov Oracle Integrations</b>   |
| <b>Compliance &amp; Support</b> | Self-Service Help Docs | Email Support              | Priority Support            | <b>Dedicated Account Manager</b> |
|                                 |                        |                            |                             | <b>Auditor Nodes (Read-Only)</b> |
|                                 | Community Support      | Standard Service Agreement | Enhanced SLA                | <b>Single Sign-On (SSO)</b>      |



## **13. Additional Details**

### **13.1 Market Analysis: The Immediate Need for Trust Infrastructure**

Despite being a ₹15+ lakh crore industry, India's logistics sector is still extremely structurally fragmented. With 6.3 crore MSMEs driving over 90% of the unorganized market, the absence of a digital trust infrastructure has serious financial and human costs. For example, poor monitoring and dispersed distribution cause an estimated 40% yearly loss in food and agricultural produce in India's cold chain and agriculture sectors. Even if the Indian cold chain business is growing quickly, physical infrastructure investments are still vulnerable to "Garbage In, Garbage Out" as long as there is no verifiable compliance. Similar, frequently lethal weaknesses exist in the pharmaceutical sector. The serious repercussions of unreliable supply chains are brought to light by tragedies such as the Gambia cough syrup tragedy and the Coldbest-PC poisoning. There is an urgent market need for completely traceable, tamper-proof pharmaceutical logistics due to growing regulatory pressure from CDSCO and international organizations. As a result, CryoTrack is positioned not only as a software platform but also as the fundamental digital public infrastructure (DPI) for this shift, reaching a sizable Total Addressable Market (TAM) that is in dire need of reasonably priced compliance solutions.

### **13.2 User-Adoption Strategies: Zero-Friction Scaling**

Hardware limitations and tech knowledge are more important issues in Indian logistics than software availability. The goal of CryoTrack's go-to-market strategy is to eliminate all obstacles to entrance for entry-level employees. CryoTrack can also be used as a Progressive Web App (PWA) as part of our mobile-first approach. Downloading a hefty native program is a major source of friction for the lowest-infrastructure actors, such tiny storage owners or rural farmers. By providing a lightweight, browser-based interface that works like our native app with some reduced capabilities, the PWA gets around this. Users can send location pins, and confirm handoff liabilities with ease. Importantly, the PWA is designed for low-end Android devices with sporadic connectivity; an offline caching mechanism guarantees that users in deep cold storage units or remote agricultural belts can log checkpoints without experiencing network frustration, securely anchoring inputs to the ledger once connectivity is restored. In addition to the PWA, CryoTrack incorporates a WhatsApp bot that is only focused on sending automated status updates, essential transit alerts, and real-time notifications straight to a platform that these employees already check on a daily basis. We will work with enterprise distributors to implement a "top-down mandate with bottom-up incentives" strategy to promote initial adoption. While ground-level MSMEs would be encouraged to embrace the PWA through quicker payment clearing and shorter dispute resolution periods, enterprises will need the platform for compliance.

### **13.3 Potential Partnerships**

Commercial partnerships will fuel our transaction volume and ecosystem expansion, while government collaborations will offer the required regulatory tailwinds. Our first commercial deployment focuses on industries where current supply chain systems are either completely nonexistent (still using paper logs), dependent on easily manipulated legacy software, or fundamentally flawed. By building trust in these zero-infrastructure settings, we lay the groundwork for a far more extensive growth. This infrastructure gap offers a vital potential in the manufacturing and heavy industrial sectors, in addition to our principal concentration on cold-chain logistics. With a special "Scope 3" emissions tracking module, CryoTrack is in a unique position to assist these industrial businesses. Businesses can cryptographically verify their indirect carbon footprints across highly fragmented, multi-tiered vendor networks by using our immutable ledger to anchor transit and operational data. This capability is increasingly required by global ESG reporting standards, but it currently lacks a verifiable, tamper-proof data layer.

We are focusing on certain infrastructure collaborations to collect and safeguard ground-level data in order to carry out this development across the cold-chain and industrial sectors. Our goal is to collaborate with local IoT and hardware producers who make reasonably priced, unencrypted GPS trackers and temperature sensors. By transforming widely accessible hardware into enterprise-grade verified oracles, CryoTrack will serve as the cryptographic software layer that protects their raw telemetry data. Additionally, by combining our API Gateway with agritech and B2B e-commerce platforms like Udaan, Ninjacart, or DeHaat, we will be able to plug straight into current aggregators and instantly give severely disordered industries the layer of decentralized transit integrity that they need.

We are looking into collaborations with telecom companies like Jio and Airtel for edge-computing capabilities and secure, localized data processing to support the massive data loads of real-time industrial and Scope 3 tracking. Our Tier 2 AI Evaluation Engine's real-time capabilities for immediate transit alerts will be significantly enhanced by utilizing 5G infrastructure. Lastly, we intend to work with freight and cargo insurance firms by offering verified, WORM-enforced ledger data. Insurance claims can be resolved via smart contracts in days rather than months by providing cryptographically guaranteed proof of where a supply chain failed and precisely who was liable at that time. This greatly encourages insurers to require CryoTrack for their industrial and agricultural policyholders.

#### 14. Signature and Team Information

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